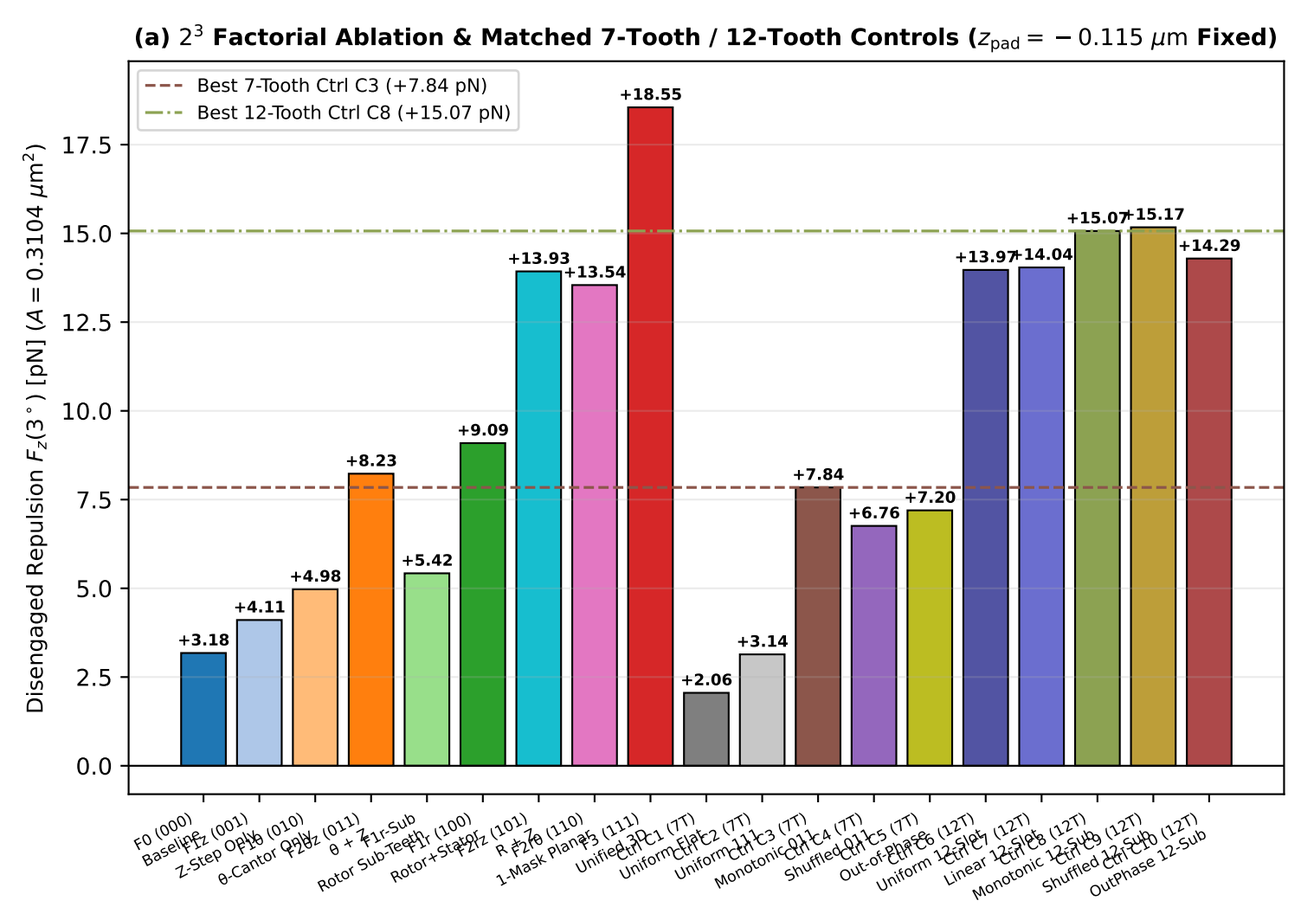

(a) 2^3 Factorial Ablation & Matched 7-Tooth / 12-Tooth Controls ($z_{pad} = -0.115$ μm Fixed)
Best 7-Tooth Ctrl C3 (+7.84 pN)
Best 12-Tooth Ctrl C8 (+15.07 pN)
Disengaged Repulsion $F_z(3°)$ [pN] ($A = 0.3104$ μm^2)
+3.18
+4.11
+4.98
+8.23
+5.42
+9.09
+13.93
+13.54
+18.55
+2.06
+3.14
+7.84
+6.76
+7.20
+13.97
+14.04
+15.07
+15.17
+14.29
F0 (000) Baseline
F1z (001) Z-Step Only
F1θ (010) θ-Cantor Only
F2rz (011) θ + Z
F1r-Sub Rotor Sub-Teeth
F1r (100) Rotor+Stator
F2rz (101) R + Z
F2rθ (110) 1-Mask Planar
F3 (111) Unified 3D
Ctrl C1 (7T)
Uniform Flat
Ctrl C2 (7T)
Uniform 111
Ctrl C3 (7T)
Monotonic 011
Ctrl C4 (7T)
Shuffled 011
Ctrl C5 (7T)
Out-of-Phase
Ctrl C6 (12T)
Uniform 12-Slot
Ctrl C7 (12T)
Linear 12-Slot
Ctrl C8 (12T)
Monotonic 12-Sub
Ctrl C9 (12T)
Shuffled 12-Sub
Ctrl C10 (12T)
OutPhase 12-Sub
(b) Per-Slot Force Breakdown Across the 7 Stator Radial Bands
F0 (000) Baseline
F1r (100) Radial Cantor
F3 (111) Unified 3D Fractal
C3 (7T) Monotonic 011
C8 (12T) Monotonic 12-Sub
Per-Slot Axial Force $F_{z,k}$ [pN]
S1 (Gen3)
S2 (Gen2)
S3 (Gen3)
S4 (Gen1)
S5 (Gen3)
S6 (Gen2)
S7 (Gen3)
(c) Wavenumber Spectrum $\kappa f_z(\kappa)$ Across (r, θ, z) Fractal Dimensions
F0 (000) Baseline
F1r (100) Radial Fractal
F2rθ (110) 1-Mask Planar
F3 (111) Unified 3D Fractal
Ctrl C3 (7T) Monotonic
Ctrl C8 (12T) Monotonic 12-Sub
Ctrl C9 (12T) Shuffled 12-Sub
Spectral Integrand $\kappa f_z(\kappa)$ [pN]
κ_1 (Gen 1)
κ_2 (Gen 2)
κ_3 (Gen 3)
Imaginary Wavenumber $\kappa = \xi/c$ [μm^{-1}]
(d) Scale-Locked Generation Spectra ($m = 1, 2, 3$) in Unified 3D Fractal vs. Baseline
F3 Gen-1 Slot (S4): 3 Cantor Sub-Teeth
F3 Gen-2 Slots (S2, S6): 2 Cantor Sub-Teeth
F3 Gen-3 Slots (S1, S3, S5, S7)
F0 Gen-1 Slot (Baseline)
F0 Gen-2 Slots (Baseline)
Generation-Resolved $\kappa f_{z,m}(\kappa)$ [pN]
Imaginary Wavenumber $\kappa = \xi/c$ [μm^{-1}]